

# Code-Layout, Readability and Reusability

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 04 July 2026                                       |
| Team ID       | SOLO-CCFD-2026                                     |
| Project Name  | Credit Card Fraud Detection using Machine Learning |
| Maximum Marks | 5 Marks                                            |

## Code Layout Checklist

| S.No | Code Quality Parameter    | Followed | Remarks                                                                                     |
|------|---------------------------|----------|---------------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Yes      | PEP8-style 4-space indentation used throughout train_model.py and app.py                    |
| 2    | Proper File Structure     | Yes      | Separated into model/ (training + deployment) and flask_app/ (web app) folders              |
| 3    | Meaningful Variable Names | Yes      | e.g. X_train_bal, fraud_probability, best_model_name                                        |
| 4    | Function / Method Names   | Yes      | load_data(), preprocess(), train_and_compare(), prepare_row()                               |
| 5    | Code Comments             | Yes      | Docstrings on every script plus inline comments explaining SMOTE, scaling, thresholding     |
| 6    | Modular Design            | Yes      | Preprocessing, training, evaluation, and saving are separate reusable functions             |
| 7    | No Redundant Code         | Yes      | Single shared preprocessing function used by both training and Flask inference              |
| 8    | Error Handling            | Partial  | Flask /predict route wraps input parsing in try/except; batch route can be hardened further |

## Reusable Components / Modules

| S.No | Component / Module          | Language/Tech | Reusability | Where Reused                                            |
|------|-----------------------------|---------------|-------------|---------------------------------------------------------|
| 1    | preprocess()                | Python/pandas | High        | Used in both train_model.py and app.py's prepare_row()  |
| 2    | StandardScaler (scaler.pkl) | scikit-learn  | High        | Reused at inference time to match training-time scaling |
| 3    | Model comparison loop       | Python        | Medium      | Reusable pattern for adding new algorithms in future    |

## Overall Code Quality Assessment

| Aspect                   | Rating (1-5) | Comments                                                    |
|--------------------------|--------------|-------------------------------------------------------------|
| Code Layout & Structure  | 5            | Clear separation of training, deployment, and web app code  |
| Readability              | 5            | Descriptive names, docstrings, consistent style             |
| Reusability              | 4            | Preprocessing logic shared between training and serving     |
| Documentation / Comments | 5            | Every script has a module docstring and inline explanations |
| Overall Score            | 4.75         |                                                             |